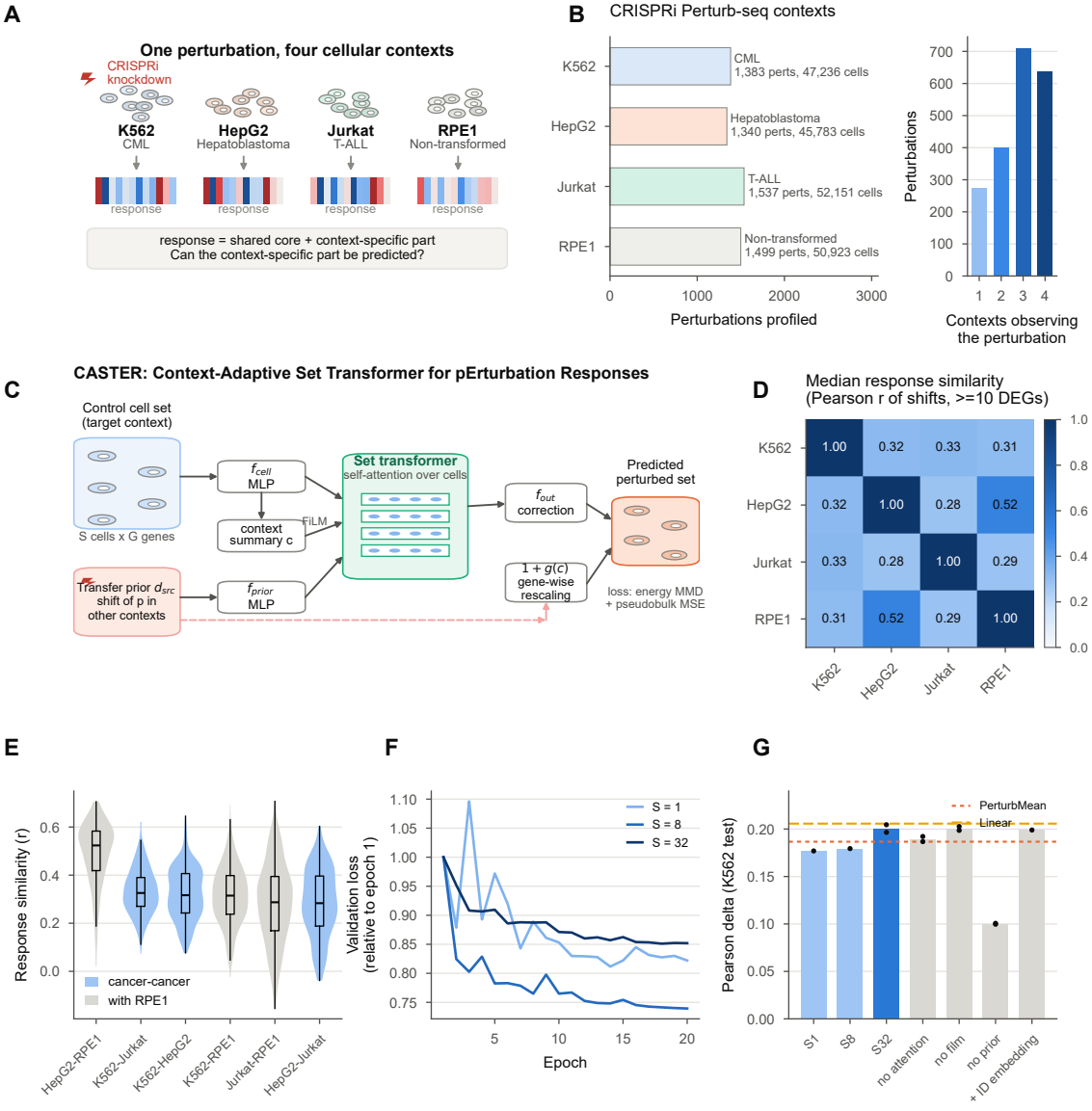

A
One perturbation, four cellular contexts
CRISPRi knockdown
K562
CML
HepG2
Hepatoblastoma
Jurkat
T-ALL
RPE1
Non-transformed
response
response = shared core + context-specific part
Can the context-specific part be predicted?
B
CRISPRi Perturb-seq contexts
K562
CML
1,383 perts, 47,236 cells
HepG2
Hepatoblastoma
1,340 perts, 45,783 cells
Jurkat
T-ALL
1,537 perts, 52,151 cells
RPE1
Non-transformed
1,499 perts, 50,923 cells
Perturbations profiled
Perturbations
Contexts observing the perturbation
C
CASTER: Context-Adaptive Set Transformer for pErturbation Responses
Control cell set (target context)
S cells x G genes
Transfer prior d_{src} shift of p in other contexts
f_{cell} MLP
context summary c
FiLM
f_{prior} MLP
Set transformer
self-attention over cells
f_{out} correction
$1 + g(c)$ gene-wise rescaling
Predicted perturbed set
loss: energy MMD + pseudobulk MSE
D
Median response similarity
(Pearson r of shifts, >=10 DEGs)
K562
HepG2
Jurkat
RPE1
1.00
0.32
0.33
0.31
0.32
1.00
0.28
0.52
0.33
0.28
1.00
0.29
0.31
0.52
0.29
1.00
E
Response similarity (r)
cancer-cancer
with RPE1
HepG2-RPE1
K562-Jurkat
K562-HepG2
K562-RPE1
Jurkat-RPE1
HepG2-Jurkat
F
Validation loss (relative to epoch 1)
S = 1
S = 8
S = 32
Epoch
G
Pearson delta (K562 test)
PerturbMean
Linear
S1
S8
S32
no attention
no film
no prior
+ ID embedding